Machine Learning Basics

A short introduction covering machine learning, neural networks, deep learning, supervised and unsupervised learning, and overfitting.

What is Machine Learning?

Machine learning is a branch of artificial intelligence in which computers learn patterns from data instead of following rules written by a programmer. A machine learning model is trained on examples, and it then uses what it learned to make predictions or decisions on new data it has not seen before.

Common uses include spam filtering, product recommendations, speech recognition, fraud detection and medical image analysis. A typical workflow is to collect data, clean it, split it into training and test sets, train a model, evaluate it, and then improve it.

What are Neural Networks?

A neural network is a machine learning model loosely inspired by the brain. It is built from layers of connected units called neurons. Each neuron takes numbers as input, multiplies them by weights, adds a bias, and passes the result through an activation function such as ReLU or sigmoid.

A network has an input layer, one or more hidden layers and an output layer. During training, the network compares its output with the correct answer using a loss function. It then adjusts its weights using backpropagation and gradient descent so that the error becomes smaller over many passes through the data.

What is Deep Learning?

Deep learning is a part of machine learning that uses neural networks with many hidden layers. The extra layers let the model learn features automatically, from simple patterns in early layers to complex concepts in later layers.

Deep learning works especially well on images, audio, video and text. Convolutional neural networks are widely used for images, recurrent networks and transformers for sequences and language. Deep models usually need large amounts of data and more computing power than traditional methods.

Supervised vs Unsupervised Learning

Supervised learning uses labeled data, where every training example comes with the correct answer. The model learns to map inputs to outputs. Classification predicts a category, such as spam or not spam, and regression predicts a number, such as a house price. Examples of algorithms are linear regression, decision trees, support vector machines and random forests.

Unsupervised learning uses data without labels. The model looks for structure on its own. Clustering groups similar items together, for example customer segments, and dimensionality reduction simplifies data while keeping its main patterns. K-means and PCA are common examples.

The main difference is that supervised learning needs labeled answers to learn from, while unsupervised learning discovers patterns without them.

How to Prevent Overfitting

Overfitting happens when a model learns the training data too closely, including its noise, so it performs well on training data but poorly on new data. The warning sign is a large gap between training accuracy and validation accuracy.

Ways to reduce overfitting include: collecting more training data; using a simpler model; applying regularization such as L1 or L2 penalties; using dropout in neural networks; stopping training early when validation error stops improving; using cross-validation to check performance; and using data augmentation to create more varied training examples.

The opposite problem is underfitting, where the model is too simple to capture the pattern in the data. Good model building aims for a balance between the two.